INCH-POUND

A-A-59355
19 October 1998

COMMERCIAL ITEM DESCRIPTION

LEAK DETECTOR, HELIUM

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This document outlines the minimum requirements for a helium leak detector. This detector will be used to calibrate mass spectrometers at military installations around the world.

2. SALIENT CHARACTERISTICS.

2.1 General. This calibrated helium leak detector consists of a controlled source of helium ion gas traceable to the National Institute of Standards and Technology (NIST) external reference standards used in calibrating leak detectors.

2.2 Leak rate. The leak rate is 10^{-7} std cc/sec minimum detectable partial pressure of helium. This factor relates to the minimum detectable leak rate in std cc/sec for helium molecules (1 std cc/sec = 0.76 torr-liter/sec).

2.3 Leak rate accuracy. The leak rate shall be less than or equal to $\pm$ 10 percent std cc/sec.

2.4 Media. The media shall be helium gas.

2.5 Type of leak. External application.

2.6 Fittings. 1-1/8-inch outside diameter compression fitting.

2.7 Value type. Manual isolation.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to SA-ALC/TILDD, 485 Quentin Roosevelt Rd., Kelly AFB, TX 78241-6425 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document.

AMSC N/A FSC 4940

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2.8 Outer casing material. Corrosion-resistant stainless steel.

2.9 Calibration. Each leak detector shall be calibrated and traceable to the NIST reference standard.

2.10 Marking. As a minimum, the helium leak detector should be permanently marked with the following information: type of calibration gas for the mass spectrometer, helium leak rate, date the calibration leak was filled with gas, calibration temperature at Celsius, model number, serial number of item and manufacturer name and address.

2.11 Instructions. An instruction and operation/maintenance manual shall be provided with the detector.

3. REGULATORY REQUIREMENTS.

3.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

4.2 Bid samples. Bid samples are required for examination and testing to determine whether the leak detector meets the requirements in this CID, provides calibration for helium mass spectrometers, and has acceptable form, fit, function and suitability for use. Offerors shall supply one (1) sample of the actual item proposed to be supplied under the government contract to SA-ALC/TIELA, Bldg. 320, 508 Shop Lane, Kelly AFB, TX 78241-6426.

5. PACKAGING.

5.1 Packaging, preservation, and packing. Packaging, preservation, and packing shall be as specified in the contract or order.

6. NOTES.

6.1 Ordering data.

a. Title, number and date of this CID.
b. Calibration report (see 2.9).
c. Instruction and operation/maintenance manual. (see 2.11).
d. Number of bid samples required (see 4.2).
e. Packaging requirements (see 5.1).

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6.2 Document ordering. Copies of federal and Department of Defense specifications, standards, and handbooks are available from the Defense Automated Printing Service, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094. Phone (215) 697-2179.

6.3 Possible Acceptable Products.

Manufacturer	CAGE	Model
Leybold Inficon Two Technology Place East Syracuse, NY 13057-0714	56507	99077072

MILITARY INTERESTS:

Custodian:
Air Force – 99
Army - AL

Reviewers:
Air Force - 01
DLA - CC

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

Preparing Activity:
Air Force – 82

Project 4940-0749

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete block 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amen contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-59355	2. DOCUMENT DATE (YYMMDD) 98101

3. DOCUMENT TITLE
LEAK DETECTOR, HELIUM FSC 4940

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	D. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. Name San Antonio Air Logistics Center/TILDD	b. TELEPHONE (Include Area Code) (1) Commercial (210)925-6314 (2) AUTOVON 945-631
c. ADDRESS (Include Zip Code) 485 Quentin Roosevelt Rd., Bldg 171 Kelly AFB, TX 78241-6524	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS CONTACT THE: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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